

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems.(L4,L5)

Assessment Tool : Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?

17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I N. Meghana (192411129) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Meghana

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

SOAP stands for Simple Object Access Protocol. It is an XML-based protocol used to exchange structured information between applications over a network. It is commonly used in web services for reliable and secure communication.

2. Define WSDL.

WSDL stands for Web Services Description Language. It is an XML-based document that describes a web service, including its operations, messages, data types, bindings, and service endpoint.

3. What is an API?

API stands for Application Programming Interface. It is a set of rules and methods that allows one software application to communicate with another application or service.

4. Define XML Schema.

XML Schema, or XSD, defines the structure, elements, attributes, and data types allowed in an XML document. It is used to validate XML data.

5. What is a Web Service?

A web service is a software service that allows different applications to communicate and exchange data over a network using standard web technologies such as HTTP, XML, SOAP, or REST.

6. Define REST.

REST stands for Representational State Transfer. It is an architectural style for building web services using resources and standard HTTP methods such as GET, POST, PUT, and DELETE.

7. What is a Protocol?

A protocol is a set of rules that defines how data is transmitted and received between communicating devices or applications. Examples include HTTP, HTTPS, TCP, and SOAP.

8. Define Endpoint in web services.

An endpoint is the specific URL or network address through which a web service can be accessed. A client sends requests to the endpoint to use the service.

9. What is a Service in web applications?

A service is a software component that performs a specific function and provides that function to other applications through an interface, usually over a network.

10. What is the difference between SOAP and REST?

SOAP is a protocol that mainly uses XML messages and follows a strict message structure. REST is an architectural style that commonly uses HTTP methods and can use formats such as JSON or XML. REST is generally simpler and lightweight, while SOAP provides built-in standards for features such as security and reliability.

11. What is the role of WSDL in web services?

WSDL describes how a web service can be used. It tells the client what operations are available, what data is required, what messages are exchanged, and where the service is located.

12. What is JSON and how is it used in APIs?

JSON stands for JavaScript Object Notation. It is a lightweight data format used to represent and exchange data. Many REST APIs use JSON because it is simple, compact, and easy for applications to read and process.

13. What is HTTP and why is it used in web services?

HTTP stands for HyperText Transfer Protocol. It is used to transfer requests and responses between clients and servers. Web services commonly use HTTP because it is widely supported and provides standard methods such as GET and POST.

14. What is meant by request and response in web communication?

A request is a message sent by a client to a server asking for data or an operation. A response is the message returned by the server containing the requested data, result, or an error message.

15. What is the importance of XML in web services?

XML provides a standard, structured way to represent and exchange data between different systems. It is platform-independent and is widely used in technologies such as SOAP and WSDL.

16. What is the purpose of an API endpoint?

An API endpoint provides the specific address through which a client accesses a particular API resource or operation. For example, an endpoint may be used to retrieve customer or product information.

17. What is the difference between HTTP and HTTPS?

HTTP transfers data without encryption, while HTTPS uses TLS encryption to protect data during communication. HTTPS is preferred for sensitive information such as passwords, payment details, and personal data.

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They represent resources using URLs and use HTTP methods such as GET, POST, PUT, and DELETE to perform operations on those resources.

19. What is a namespace in XML?

An XML namespace is used to uniquely identify elements and attributes in an XML document. It prevents naming conflicts when elements from different XML vocabularies are used together.

20. What is the role of a client and server in web services?

The client is the application that requests a service or data. The server receives the request, processes it, and sends a response. Thus, the client and server communicate through requests and responses.